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Leading Successful Digital Transformations

By Stijn Viaene

While a brain surgeon's mastery of the scalpel is key to success in the operating room, it can also be an asset when doing non-medical tasks, such a preparing sashimi for dinner. But scalpel skills offer little advantage when dealing with patient stress or changing tires or diapers. In other words, the usefulness of surgical skills depends on the context in which they are deployed. And the same can be said about some leadership skills, especially in the digital age. So why is context often overlooked when organizations seek a digital transformation leader?

What makes an effective digital transformation leader? Attempts to answer this question often rehash generic leadership qualities and training recipes. These are frequently rebranded for the purpose of digital transformation, but they still fall short.

Simply put, our research shows there is no one-size-fits-all answer to what makes an effective digital transformation leader. In this article, we propose a contingent approach to digital transformation leadership, one that recognizes the unique challenges of digital transformation and identifies a set of crucial leadership qualities that are context dependent.

This approach is based on extensive action-design research conducted by the author and his team between 2014 and 2024, which facilitated the creation of a pioneering open-enrollment executive education program on digital transformation leadership at Vlerick Business School.

DEFINING THE END GAME

We view digital transformation as a process of organizational change that leverages digital technology. It is a response to shifts in an organization's business environment driven by a continuous stream of digital opportunities and challenges. We deliberately frame digital transformation not as a tactic, but as a strategic, holistic endeavour—an end-to-end effort that spans the entire value chain and is both inward- and outward-looking.

What distinguishes digital transformation from other technology-enabled organizational transformations is its continuous nature (Vial 2019; Hanelt et al. 2021; Verhoef et al. 2021). Encompassing constant shifts from one technology architecture to another, it results in successive waves of transformation rather than discrete projects or programs. The latest wave, driven by AI, is reshaping how humans interact with technology, pushing the boundaries of autonomous decision-making and work (Brynjolfsson 2022; Duan, Edwards, and Dwivedi 2019; Leonardi 2023). Digital technology plays a dual role in the process, as digital transformation is about organization-wide change both in response to and with the help of digital technologies.

So, what makes digital transformations effective and how do we judge their success? The answer lies in the objective. In the past, technology transformations were primarily about optimizing costs and efficiency and reducing variance. But today's dynamic business environment demands that digital transformations aim for a different goal while still ensuring optimal cost and efficiency, and low variance. This goal is organizational agility (Tallon et al. 2019; Viaene 2020; Teece, Peteraf, and Leih 2016), which is the ability to routinely explore, and exploit, opportunities presented by digital technologies faster than competitors. The priorities are speed to market, flexibility, and responsiveness.

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In addition to being the goal, organizational agility is the measure of success. Yet, pinpointing the exact level of agility required is not straightforward. It depends on what digital technologies can offer and what these technologies require from an organization, again reflecting their dual role in digital transformations as both enablers and drivers of change.

LEADING FOUR TRANSFORMATION CONTEXTS: VIGILANT, VOYAGER, VISIONARY, AND VESTED

Regardless of whether organizations take a top-down or bottom-up approach, digital transformations need champions who can guide their organizations into a future characterized by digital-age organizational agility. The effectiveness of digital transformation leaders is measured by their ability to introduce and foster good new practices that operate at digital speed and scale.

But introducing new, agile ways of working is different than developing strategies or introducing skills, and what is considered a "good practice" depends largely on the specific context in which a leader operates. An approach that is effective in one context may not yield the same results in another. Similarly, the benchmarks for success may differ across contexts. For example, while ROI is a conventional metric for assessing the financial benefits of implementing a new ERP system by Finance & Accounting, it is short-sighted, if not counterproductive, to use it to evaluate a hackathon launched by R&D to develop AI systems for cross-species communication.

As an organization-wide phenomenon, digital transformation inevitably spans multiple contexts within an organization. We identify four primary contexts for studying digital leaders within organizations undergoing such transformation. Each context is defined by two key dimensions of organizational tension—transformational purpose and scope—that influence the digital transformation mandate, as well as the criteria for judging its success.

The first dimension, transformational purpose, involves a tension—exploiting versus exploring (Bughin et al. 2018; March 1991). Transformational purpose qualifies whether progress with the introduction of a new way of working by digital transformation leaders is sought in the space of the organization's exploration agility or exploitation agility. While exploitation and exploration require different structures, cultures, and processes, there is a large body of research and evidence that shows that organizations need to balance and leverage both for a successful digital transformation (Andriopoulos and Lewis 2009; Danneels and Viaene 2022; Smith and Beretta 2021; Smith and Lewis 2011). The ability to effectively transform the organization's exploitation and exploration is critical to organizational agility and long-term success.

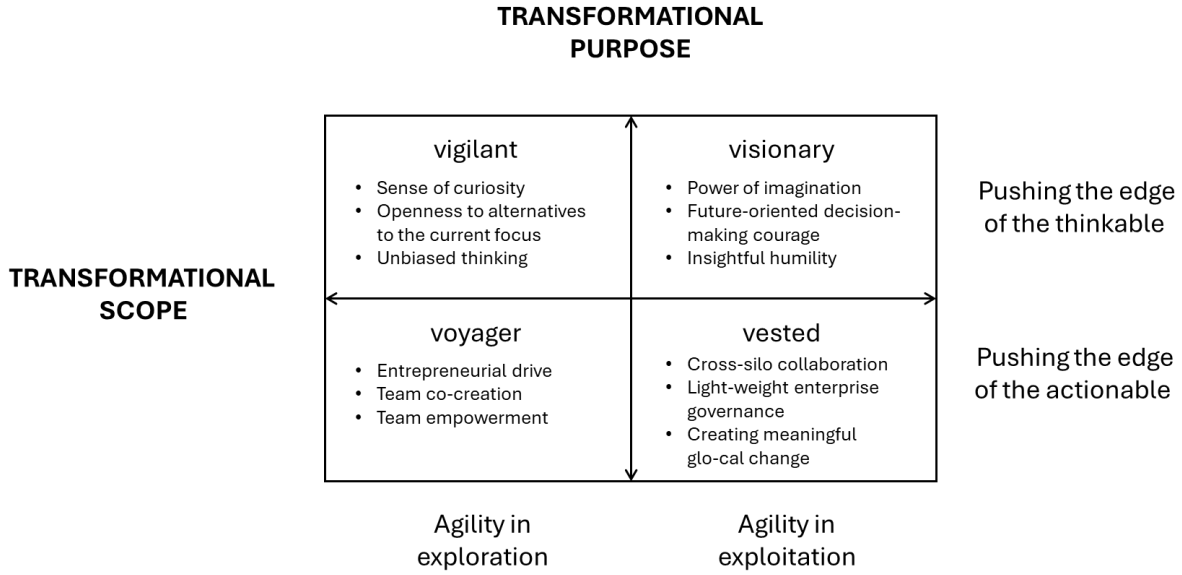
"Regardless of whether organizations take a top-down or bottom-up approach, digital transformations need champions who can guide their organizations into a future characterized by digital-age organizational agility."

The second dimension, transformational scope, involves another tension—thinking versus action. Transformational scope qualifies the latitude afforded to digital transformation leaders in seeking to push the edge of the thinkable or the actionable with the introduction of new, digital-savvy agile ways of working. Making sense of digital opportunities and challenges requires stimulating both thinking and acting differently, but sense-making activities may exhibit a drive biased towards one or the other, as experienced by practitioners and reflected conceptually (Choo 2002; Weick 1995). While it is important to think big and demonstrate the audacity of thought, it is equally important to balance this with pragmatism to ensure digital transformations are implementable and can lead to real, tangible change.

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Plotting these two dimensions of organizational tension on a pair of non-quantitative axes produces four distinct quadrants, each corresponding to a context (see FIGURE 1) in which the notion of effectiveness or success is different. For example, the previously mentioned implementation of a new ERP system is a typical example of a digital transformation initiative executed in a "vested" context, whereas the hackathon aimed at developing AI systems for cross-species communication typically finds its place in a "vigilant" context.

FIGURE 1: DIGITAL TRANSFORMATION CONTEXTS AND ROLE-MODEL LEADER QUALITIES



Our research investigated the behaviours and practices of leaders in each of the digital transformation contexts identified above looking to identify leadership qualities that appear conducive to achieving success. While we found some overlap, certain leadership behaviours and practices were more prevalent in specific contexts. From these prototypical behaviours and practices, we inferred the qualities that effective digital transformation leaders tend to role-model, depending on the context in which they primarily operate.

ROLE-MODEL QUALITIES IN THE VIGILANT CONTEXT

Digital transformation leaders active in a vigilant context are inclined to respond to their organization's changing business environment by staying alert and monitoring and anticipating the upcoming changes. As a result, their main role-model qualities are a sense of curiosity, openness to alternatives to the current focus, and unbiased thinking.

Sense of curiosity: Vigilant digital transformation leaders are driven to grasp the organization's contextual turbulence, which is catalyzed by digital technologies. They actively seek out a wide variety of cues, signals, ideas, and perspectives utilizing digital technologies to stay alert. They are committed to learning about technological advancements, emerging trends, and shifts in the broader organizational environment. They remain attentive to potential opportunities and threats, extending their focus and scope beyond their organization, business model, or industry. The vigilant leaders' curiosity involves broad and in-depth exploration of the technological and business context.

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Openness to alternatives: Vigilant digital transformation leaders pay particular attention to creating an atmosphere of openness for and making sense of perspectives alternative to the current beliefs or assumptions. They study current organizational biases and assumptions to be able to tether alternative views in a meaningful way to the organizational decision-making or action frames others currently live in and might be trapped in. Vigilant leaders prefer to collectively craft these alternative perspectives to increase their organizations' preparedness for accepting and entertaining different ways events could develop.

Unbiased thinking: Vigilant digital transformation leaders take a broad and long view of possible contextual developments. They strive to abductively come to terms with a rich variety of uncertainties, weak signals, emerging and other trends, and ideas to enable others to logically engage with alternative plausible scenarios for environmental developments and new possibilities for conducting business and organizing work in the digital age. They create a context that suspends judgment and allows critical argumentation for several believable interpretations of how the organization's environment might unfold. Beliefs, assumptions, biases, and personal preferences are made explicit to allow for the exploration of their implications, including the cases in which they do not hold true.

ROLE-MODEL QUALITIES IN THE VOYAGER CONTEXT

For digital transformation leaders operating in a voyager context, it makes the most sense to respond to changes in their organization's business environment in a hands-on, entrepreneurial way. This is why entrepreneurial drive, team co-creation, and team empowerment are their three stand-out qualities.

Entrepreneurial drive: Voyager digital transformation leaders contribute to their organization's success by developing resourceful ways to make innovative digital-savvy ideas tangible. They aim to evaluate the business potential of ideas by showing value-in-use and emphasizing the value of learning early and often. They focus energy on taking initiative and skillfully navigating the challenges of starting and growing opportunities, under turbulent or adversarial conditions. They take advantage of the continuous stream of digital technologies to pragmatically innovate faster, cheaper, and better.

Team co-creation: Voyager digital transformation leaders nurture collective creativity within teams by skillfully combining diversity and creativity. They shape a team's ability to tackle challenges creatively by championing inclusion, fostering open debate, and encouraging productive conflict—all grounded in a foundation of trust. Furthermore, they instill a creative solution exploration logic that values iterating on problems and solutions, swiftly adapting to changes, and learning from failure. Voyager leaders create high-performance teams that embrace change and thrive on it.

Team empowerment: Voyager digital transformation leaders entrust teams with high degrees of freedom to decide how to act. This autonomy, inspired by the need for speed, flexibility, and responsiveness, encompasses teams taking responsibility and being accountable for solution-scoping decisions (involving desirability, feasibility, and viability) and shaping their (team) work organization and processes. Voyager leaders ensure that this team empowerment can work by creating a high-trust environment where organizational stakeholders respect team decisions and by clearly framing the team's autonomy within a clearly articulated broader organizational value frame.

ROLE-MODEL QUALITIES IN THE VISIONARY CONTEXT

Digital transformation leaders working in a visionary context respond to the changing business environment by rethinking the purpose and capabilities of their organizations in light of the changes ahead, creating an inspiring and aspirational vision, and bringing on board stakeholders from inside and outside their

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organizations. This makes power of imagination, future-oriented decision-making courage, and insightful humility their key qualities.

Power of imagination: Visionary digital transformation leaders, inspired by their organizations' need to survive and thrive in the digital age, imagine and frame their organizations' purpose and winning formula. Taking an ecosystem perspective of strategy and customer value creation, they envision a to-be future for the organization that benefits from engaging in digital win–win partnerships. Against the backdrop of an always turbulent environment, they construct and communicate a believable, engaging, and energizing story of the organization transforming and thriving with a digital savvy mindset.

Future-oriented decision-making courage: Visionary digital transformation leaders make bold choices. They think in terms of informed strategic bets and real options activating new growth paths, implying both radical evolution of the current core and creation of new adjacent or self-disruptive growth paths. These leaders do not shy away from addressing decision paradoxes. Thus, solving the puzzle of finding a meaningful relationship between people and technology is intrinsic to their motivation. Their pulling a growth mindset card for digital transformation implies choosing to grow current and new digital business by cultivating and investing in learning and growth in people. This choice allows visionary leaders to frame transformation and change as positive, emotional, and empathic. It also drives them to instill their visions of digital transformation with a sense of purpose.

Insightful humility: Visionary digital transformation leaders not only understand the digital landscape and the organizations' future needs but also approach their leadership with a humble mindset. Acknowledging their own limitations, they are open to continuous learning, seek and embrace evidence and honest feedback from diverse perspectives, and are willing and able to consider adapting their strategies in an ever-evolving digital environment. Their credibility rests on their ability to balance strong advocacy for stable purpose, vision, and transformation principles and care for an inclusive and collaborative approach to emergent decision-making.

ROLE-MODEL QUALITIES IN THE VESTED CONTEXT

Digital transformation leaders involved in a vested context are primarily concerned with harnessing the potential of digital technologies to operationalize the newly chosen value pathways to win at scale. They can, therefore, be identified by their commitment to (1) cross-silo collaboration, (2) lightweight enterprise governance, and (3) creating meaningful global change.

Cross-silo collaboration: Vested digital transformation leaders aim to shift the agile exploitation frontier of digital transformation at scale by establishing an organization that fosters fluid and productive cross-boundary connections and collaborations among individuals. This network organization, inclusive of internal and ecosystem work, is orchestrated by a robust common sense of customer value. Empowered by digital technology, these leaders are dedicated to removing bureaucratic hurdles, breaking down functional silos, and improving information flows to minimize delays in decision-making and action. This commitment extends both vertically within hierarchies and horizontally across value chains, supply chains, and other ecosystem collaborations.

Lightweight enterprise governance: Vested digital transformation leaders move the organization beyond visioning and experimentation by steadily and progressively working towards a more productive, flexible, and responsive organizational whole. Instead of focusing on individual aspects or pieces, they map out a course for the entire enterprise to succeed with digital transformation. Their digital target operating model represents a vision of a composable enterprise made up of conveniently combinable and reusable digital component assets. These architects of enterprise change with a bias for action adopt a lightweight

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framework for orchestrating and governing change that effectively aligns objectives, coordinates resource allocation, and integrates transformation outputs across the enterprise without stifling swift strategic action.

Creating meaningful glo-cal change: Vested digital transformation leaders strive to engage the entire organization in digital transformation by ensuring that required changes hold significance and benefit both individuals and the organization. They focus on establishing work environments that motivate and support participants to learn and adopt common digital technologies, work practices, and organizational methods. They are committed to community-building, emphasizing empowerment, collaboration, shared values, and a collective sense of purpose and learning among its members. Vested leaders effectively guide the organization in establishing the 'global' vision of a digital transformation inspired by a sense of purpose and rooted in a growth mindset a 'local' reality. They are actively engaged in removing psychological and material obstacles with a mindset geared to empower workers.

TOWARDS CONTINGENT AND INCLUSIVE LEADERSHIP

Our research highlights why the traditional one-size-fits-all approach to leadership development is not appropriate for digital transformations. What does this mean in practice? For one thing, it means that a deep sense of curiosity and openness to alternatives to the current focus is not necessarily an asset for digital transformation leaders. In fact, these qualities could potentially lead some project teams off track.

Moreover, since digital transformation leaders may find themselves working in different contexts over time, they need to understand that being effective in one context does not guarantee success in another. They need to hone different qualities as they move from one project to the next.

Digital transformation is not driven by a single omni-qualified digital-savvy individual, which is why investing in leadership training focused on all the role-model qualities noted above is pointless. In addition to a decoupling from top management, success calls for the development of champions who can guide the introduction of new, agile ways of working, and the qualities they need depend on the specific context in which they work. That is why a customized leadership development journey that focuses on context-relevant qualities is the best path to digital transformation success.

An important corollary, however, is that these leaders, working in their respective contexts, should not act to the detriment of transformation activities in other contexts, lest the holistic, end-to-end nature of digital transformation be compromised. This not only has implications for organizational governance but also points to a single quality that all digital transformation leaders should have in common, regardless of the context in which they work: a collaborative mindset.

Furthermore, since the approach we propose calls for a diverse community of digital transformation leaders endowed with context-dependent qualities, it makes a powerful case for diversity and inclusion. This expands an organization's leadership capacity and gives more people the opportunity to realize their leadership potential. By framing digital transformation as an open invitation, you can unlock the full potential of your organization's talent pool, including talent that may have been overlooked.

BIO: Stijn Viaene is a full professor at Vlerick Business School and at KU Leuven. As a management and technology expert, he helps organizations create productive and meaningful digital change by connecting digital transformation, organizational agility, and leadership. Contact: stijn.viaene@vlerick.com.

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